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Team Number : 39

Team Name: BEYONDERS

**Title: Leveraging LLM and RAG for Automated Answer Script
Evaluation**

Video link/QR code -2 mins video of your work



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Problem Statement:

Manual evaluation of answer scripts is time-consuming and inconsistent. This project automates the process using a fine-tuned LLM and RAG pipeline, aiming for scalable, unbiased, and efficient grading through AWS SageMaker.

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Introduction



The conventional approach to answer script evaluation is both time-intensive and susceptible to bias. This project addresses these challenges through the implementation of a fine-tuned Llama2 model integrated with a Retrieval Augmented Generation (RAG) pipeline, facilitating automated and context-driven grading.

The system is structured to provide comprehensive interfaces for students and instructors, enabling real-time feedback, grade review requests, and textbook content management. Developed using ReactJS, Django, and MongoDB, the application is deployed on AWS SageMaker, ensuring scalability and robustness. Error monitoring and performance analysis are managed through Grok, reinforcing the system's reliability.

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OBJECTIVES



- **Develop a fine-tuned LLM model for answer script evaluation.**
- **Implement a RAG pipeline to provide contextual analysis.**
- **Develop a dashboard for students and teachers to review marks.**
- **Deploy the solution using AWS SageMaker for scalability.**
- **Enable students to view detailed feedback based on the evaluation, enhancing transparency and learning outcomes.**
- **Allow teachers to adjust grading criteria based on textbook content, enabling flexible assessment approaches.**

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Solution Approach

Methodology:

- **Data Collection:** Dataset curated from Operating Systems course questions and answers.
- **Model Development:** Fine-tuning Llama2 using QLoRA for parameter-efficient tuning.
- **RAG Implementation:** Context extraction using FAISS and MiniLM embeddings.
- **Deployment:** Integration with AWS SageMaker and Lambda functions.

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Solution Approach



Key Features:

- **Contextual Grading:** Integrates Llama2 with RAG to align student responses with textbook content, enhancing grading accuracy.
- **Dual Interfaces:** Separate dashboards for students and teachers, enabling grade review requests, feedback analysis, and content uploads.
- **Real-Time Feedback:** AI-generated feedback is displayed alongside corrected scripts, providing actionable insights for students.
- **Scalability and Reliability:** AWS SageMaker handles large-scale grading requests, while Grok monitors errors and system performance.
- **Secure Data Handling:** MongoDB stores data securely with structured schemas for efficient data retrieval and updates.

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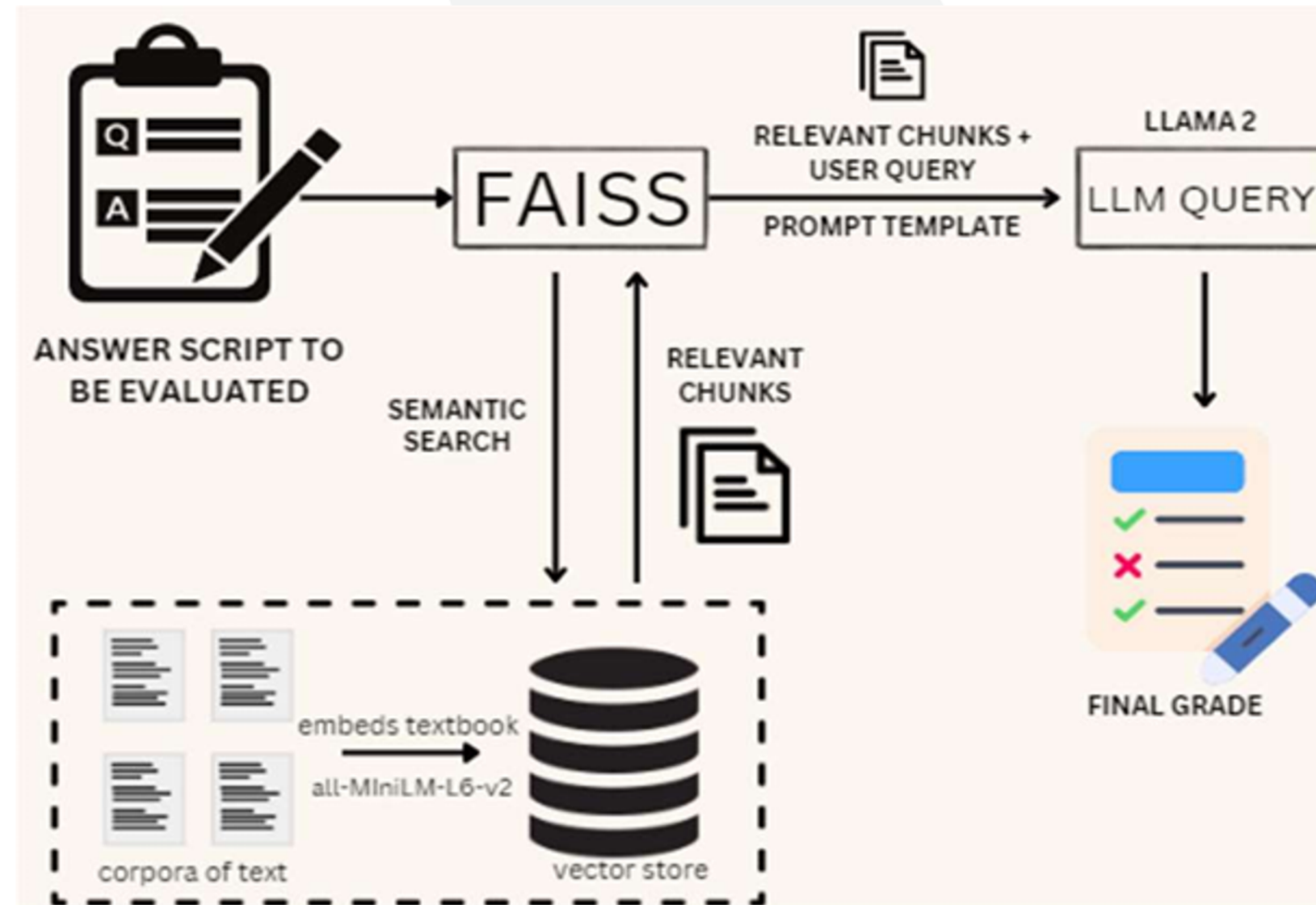
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Solution Approach

Technical Architecture:



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TOOLS & TECH USED

1. **Foundation Model:** Llama2 with Quantized Low-Rank Adaptation (QLoRA)
2. **Vector Database:** Facebook AI Similarity Search (FAISS) for efficient content retrieval
3. **Cloud Infrastructure:** AWS SageMaker for model hosting and AWS Lambda for serverless computing
4. **Database Management:** MongoDB for flexible schema document storage
5. **Frontend Development:** ReactJS framework for responsive user interfaces
6. **Backend Architecture:** Django framework for robust API development
7. **Embedding Technology:** MiniLM-L6-v2 for semantic representations

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IMPLEMENTATION

- **Successfully integrated Llama2 model with Retrieval-Augmented Generation methodology**
- **Developed comprehensive interface solutions:**
- **Faculty dashboard with evaluation metrics and review functionality**
- **Student portal with performance analytics and feedback mechanisms**
- **Implemented MongoDB database architecture for maintaining data integrity**
- **Configured AWS Lambda services for event-driven operations**
- **Established secure API communication between ReactJS frontend and Django backend**

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INNOVATIVE SOLUTION

Our solution integrates a fine-tuned Llama2 model with a RAG pipeline to enable context-aware grading, aligning student responses with textbook content for accurate evaluation.

The system provides dual interfaces for students and teachers, featuring graded scripts, AI-generated feedback, and grade review requests. AWS SageMaker ensures scalability, while Grok monitors performance and error tracking in real-time.

This approach not only automates grading but also fosters transparency and continuous feedback, enhancing academic integrity.

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SCOPE AND RESULTS

The evaluation system was rigorously tested using a dataset of 116 answer scripts from Operating Systems assessments. Key findings and insights include:

- The system clearly presents results through structured dashboards, displaying CIE and SEE scores alongside AI-generated feedback. The feedback is contextually aligned with textbook content, ensuring transparency in the grading process.**

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SCOPE AND RESULTS

- **Initial testing indicates a grading accuracy of approximately 90%, with minimal discrepancies between AI-predicted grades and teacher-assigned scores. Performance analysis reveals that the Llama2 model, fine-tuned using QLoRA, effectively handles semantic analysis, especially in short-answer questions.**
- **Analysis of grading patterns highlights potential areas for further refinement, particularly in handling ambiguous or multi-part responses. The RAG pipeline ensures logical alignment between the student's answer and the reference content, validating the scoring mechanism.**

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THANK YOU

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